Mohammed Gamal Mohammed

AI Engineer

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LinkedIn | GitHub | Portfolio

Summary

AI Engineer with hands-on experience in machine learning, deep learning, NLP and applied AI systems. Built predictive models, neural networks and information retrieval tools; contributed computer vision components to robotics and online exam proctoring projects. Skilled in Python, TensorFlow, Scikit-learn and AI model evaluation.

Education

Helwan International Technological University (HITU) | 2022–2026
Artificial Intelligence Department · Relevant coursework: Applied Deep Learning, Python for AI

Selected Projects

ISH — Intelligent Spherical Home (University Graduation Project) | 2026 GitHub-link

- Contributed to camera-based face detection and recognition for a team-built smart-home robot integrating AI, IoT and voice interaction.
- Worked on visual perception integration with the broader robotics and monitoring platform.

Technologies: Python, Computer Vision, InsightFace, FastAPI (team platform), ESP32, IoT

EduGen — AI Exam & Proctoring Platform (DEPI Graduation Project) | 2026 GitHub-link

- Developed a computer vision-based exam proctoring component for a student and instructor platform.
- Flagged suspicious left/right movements, extra people entering the camera frame and students leaving view; generated warning alerts.

Technologies: Python, Computer Vision, Camera Monitoring, Warning Logic

Medical Assistant Robot (Academic Team Project) | 2023–2024

- Contributed to a symptom-based preliminary assessment system using Decision Tree, Naive Bayes and Random Forest approaches.
- Participated in Arduino robot hardware integration, ultrasonic-sensor obstacle avoidance, testing and project documentation.

Technologies: Machine Learning, Arduino Uno, Robotics, Ultrasonic Sensors, Decision Tree, Naive Bayes, Random Forest

Training & Certifications

Generative AI Professional — Digital Egypt Pioneers Initiative (DEPI) | Nov 2025 – Jul 2026
Completed — MCIT, Egypt | Certificate of Achievement available on portfolio.

Technical Skills

Programming & Software: Python, Java, JavaScript, OOP; Flask, HTML, CSS, PHP

AI / ML: Scikit-learn, TensorFlow, Keras, supervised & unsupervised learning, evaluation, preprocessing

Applied AI: NLP, TF-IDF, information retrieval, computer vision, face recognition; familiarity with RAG concepts

Data & Systems: Pandas, NumPy, Matplotlib, Seaborn, Plotly, MySQL, Arduino, ESP32, IoT, Apache Spark

Machine Learning & NLP Projects

Neural Networks & Deep Learning Analysis (MNIST) | **Dec 2025** GitHub-link

- Trained a neural network with over 97% MNIST classification accuracy; compared SGD, Momentum, Adam and AdamW.
- Applied Dropout, L2 and early stopping; analyzed accuracy and loss curves.

Technologies: Python, TensorFlow, Keras, NumPy, Matplotlib

Loan Approval Prediction System | **Dec 2025** GitHub-link

- Built a classification pipeline with preprocessing and GridSearchCV, reporting over 85% prediction accuracy.
- Evaluated classification performance using precision, recall and confusion matrices.

Technologies: Python, Scikit-learn, Pandas, GridSearchCV

Information Retrieval Engine & Flask Interface | **Nov 2025** GitHub-link

- Built an NLP retrieval system using TF-IDF and cosine similarity to rank over 2,000 documents.
- Created a Flask interface and evaluated relevance using precision, recall and F1-score.

Technologies: Python, NLP, TF-IDF, Scikit-learn, Flask

Additional Projects

IoT Weather Prediction & Smart Control | Nov 2025 · ESP32 sensor monitoring and Naive Bayes-based control decisions; built dashboard views.

Sensitive Data Chat Monitor | Apr 2025 · Chat interface for sensitive-keyword detection, supported by Flask and Node.js. GitHub-link

Wholesale Customer Clustering | Apr 2025 · Used K-Means, DBSCAN and PCA to analyze customer segments. GitHub-link

Machine Learning Models Evaluation | Dec 2024 · Compared classification, regression and clustering results. GitHub-link

K-Means from Scratch · Implemented K-Means with multiple initialization and distance options. GitHub-link

Additional Courses

Frontend Development — Udemy (2024) · Java — Udemy (2023) · Machine Learning — Udemy (2022) · Python — Udemy (2022)